

III-V nanostructures for lasers and quantum emitters

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Abstract Selective area epitaxy is used to grow III-V semiconductor nanostructures – ranging from nanowires to nanosheets to microrings. The nanostructures are then used for various light emitting applications. ©2026 Hoe Tan

Group III-V semiconductors have revolutionised electronics and optoelectronics due to their superior physical and optoelectronic properties including high carrier mobility, direct bandgap and band structure engineering capability. Reducing the device size to nanoscale brings many unique properties, such as large surface-area-to-volume ratio, high aspect ratio, carriers and photons confinement effect. In this talk I will present our III-V research activities at The Australian National University, focusing on (i) quantum dots in nanowires and (ii) shape engineering.

(i) Incorporating single quantum dots in nanowires can enhance the spontaneous emission rate from the quantum dots for single photon emitter applications. To incorporate only one quantum in a nanowire, we utilise a combination of droplet epitaxy with selective area epitaxy to deterministically grow a InAsP single quantum dot in an InP nanowire. By optimizing growth and nanowire cavity design, single photon emission can be effectively coupled with the dominant mode of the nanowire to realize Purcell enhancement with a single-photon purity of $g^{(2)}(0)=0.05$ [1]. We further demonstrate electrically operated single photon sources from these devices [2].

(ii) Our work on selective area growth of III-V nanostructures shows the possibility of obtaining other functional nanostructures beyond the limitation of rod-like nanowires and opening the way to more advanced device geometries, such as nanosheets and microrings, with atomically smooth facets [3]. I will present our results on:

- microring lasers operating in the whispering gallery mode that can be used as on-chip lasers [4],
- coupling of a nanowire with a microring laser to control the emission direction [5],
- PT-symmetry coupled microring lasers for mode control [6] and
- nanosheets surface emitting lasers operating in the quasi-bound states in the continuum with near-unity spontaneous emission factor [7].

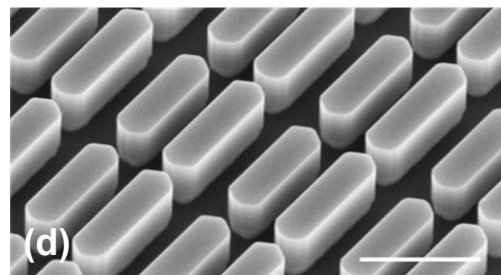
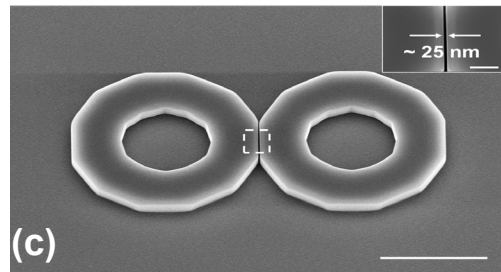
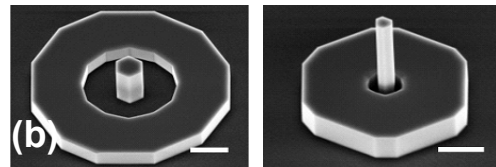
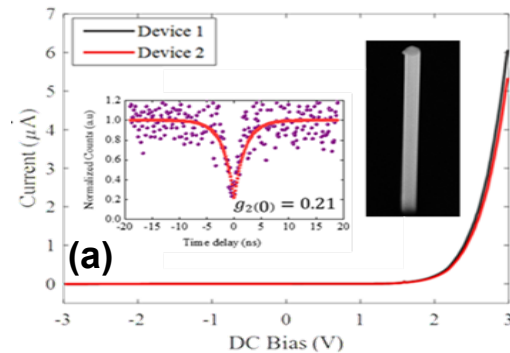


Fig. 1: (a) Electrically injected quantum dot in nanowire single photon emitter. (b) Coupled nanowire-microring lasers for directional emission. (c) Coupled pairs of microrings for lasing mode management with PT-symmetry. (d) Asymmetric nanosheet pairs for quasi-bound state in the continuum surface emitting lasers.

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